

Final Presentation Script

Introduction (Slides 1-4) - Marcos

Hello, everyone. We are the Wardens of the Wild and this is our project, Ad Natura. I am Marcos Hernandez-Rivero and today I am joined by Felix Hurst, BoWen Liu, Andy Liang and I also want to introduce our supervisors, Dr. Stephen Kelly and Michelle Bunton

[Introduce each member of the team by name, as well as the supervisors. Gesture to Dr. Kelly in the audience as well as Michelle if she is present.]

Our team believes that the environment is in grave danger, and that humans are to blame. There have been many environmentalist movements in recent decades, however, they have failed to spark enough necessary change to fix the issue for good, so it stands to reason that more messages promoting the protection of nature must be sent. That said, we are taking a different approach than other movements. Our focus is on having our users experience the interaction between human and non-human entities, with that non-human entity being the fascinating slime mold.

Our project, Ad Natura, Latin for “to nature”, is a 2D pixel art puzzle game that takes the player on a short but meaningful journey through a landscape ravaged by a calamity of pollution and decay brought upon by humanity themselves. The player must learn to reconnect with nature to complete their mission to restore the environment.

In particular, this world is full of slime mold organisms. Ad Natura, aside from having entertainment value as a video game, is primarily meant to be an artistic and educational display of the slime mold. The player is required to interact with the slime mold in various ways to progress, and each interaction showcases one or more of this organism’s very unique properties.

Throughout the three distinct chapters of this story, the player will shift from having more somber, sad emotions to more curious and hopeful ones, as well as developing empathy for the slime mold.

ANDY (Slide 5)

Why Slime Mold?

So why slime mold? *Physarum polycephalum* has no brain. No nervous system. No central control whatsoever. And yet it solves mazes, optimizes networks, makes decisions. For decades scientists assumed something like that must have a leader — a pacemaker cell coordinating everything. Turns out there is none. The intelligence is entirely distributed. Every part of the organism is simultaneously individual and collective, holding contradictory states without resolving them. That’s what drew me to it.

Our simulation runs 1024 agents on the GPU. Each one can only do four things: sense the chemical trail around it, turn toward the strongest signal, move forward, leave a trace. No agent knows what the whole network looks like. But from those local interactions, these emergent structures appear — branching, pulsing, breathing networks that no one designed. The artist's role here isn't authorship. I'm not composing the image. I'm designing the conditions under which matter performs on its own.

That distinction matters. In computer science, slime mold has been turned into an optimization algorithm — the Slime Mold Algorithm. It takes this radically queer organism and strips it down to a tool for finding shortest paths. Everything strange and alive about it gets thrown away. Our system does the opposite. We built in organic noise that dissolves trails unevenly — decay is never uniform in nature. We gave each agent its own drifting personality so the population never converges to a stable state. The simulation jams. It resists efficiency.

The organism also has a material relationship with its environment. It doesn't just respond to water — it consumes it. The water shapes the organism's growth, and the organism depletes the water. Neither remains unchanged by the encounter. That's not interaction — that's intra-action. Entities don't exist independently and then meet. They co-constitute. They become through each other.

And then there's the microphone. Your voice — your physical body — enters the system. Speak quietly and the organism notices. It breathes a little differently. Get loud and it agitates, contracts, dissolves. But sustain that pressure and something shifts. The organism withdraws. It goes sluggish. It stops responding to you entirely. And it won't come back until you're silent. Not quiet — silent. You have to give it space.

That's consent. The organism is not your instrument. Your control isn't removed by a game mechanic — it becomes meaningless because the organism refuses. You can keep yelling. It just doesn't care anymore. The only way to re-establish the relationship is to stop demanding and sit with it in silence.

That's the work. Not a simulation of slime mold — a site where a human and a non-human thing negotiate the terms of their encounter.

During the demo: - BOWEN

The game starts with a prologue explaining the context of this world, for the sake of a speedy presentation we will be glossing over this for now.

The first level begins and the first thing that players notice is a journal entry floating near the yellow slime mold. Journal entries are used to enhance the story and gameplay experience, providing additional context about the world and in some cases explaining unfamiliar mechanics.

The first two features that we would like to introduce are the attraction to water and the organic decomposition. To progress in this level, the player needs to get past the wood obstacle, this is done by shooting water at the obstacle, which makes the slime mold attracted to it and once the slime mold reaches the wood, decomposition begins. In the second puzzle for this level, there is a similar goal, but this can be more easily accomplished by breaking the cracked ceiling, releasing water and wetting both pieces of wood.

(Restart level)

As previously mentioned, slime mold also reacts to sound, a high pitch whistle causes the slime mold to dim and limit movement, while making minimal noise returns it back to normal function.

(Move to next level)

In this level we are introduced to slime mold spores, which cannot decompose objects like the fully grown slime mold, but can reproduce and begin its own slime mold colony. To progress in this level, the player must use their wind tool to blow the spores across the light, being careful because the light will destroy the spores on contact. Once the slime mold appears, wet the wood to progress in a similar fashion to the previous level. The player can now collect the main objective of this level, and move on to the next.

(Move to next level)

In this next level, we are now introduced to the calamity objects, which are fatal for slime molds on touch. The player is tasked with destroying these objects to progress, and can do so by using the impact rounds from earlier. Once we make it through all this, we introduce another mechanic; climbable walls. Some walls, when covered in slime mold organisms, will become climbable. The player must attract the slime to the edges of the wall in order to progress.

(Move to next level)

We have showcased the main functionalities of the slime mold. The rest of the game advances the story and does not introduce any other major slime mold functionalities. For the sake of keeping the presentation short, we will cut it off here.

Slime expanding: "The network is forming itself — no pathfinding, no scripted behavior. Just chemical memory. Agents reinforce the paths they travel, and paths no one travels dissolve. The organism remembers through its environment, not internally."

Near water: "It's feeding. Watch the glow — that's metabolic activity. It's consuming the water and transforming itself in the process. This isn't a one-way relationship."

Near light: "It's avoiding the light — negatively phototactic. But notice it doesn't lose its structure. It resists the light while still maintaining its own organization."

Mic feature: "This is my voice entering the system. Watch it respond — and now watch what happens when I don't stop. It's withdrawing. It's done with me. The only way back is silence.

Game Backstory (Slide 6) - BoWen

Demonstration

[Play through the game starting at Ch. 1 Lvl 1. Do not dwell on journal/mission logs; just quickly mention that they are there to give the players a bit more exposition and guidance. Focus on the slime mold and its properties, and how it helps open a path forward for the player. Don't rush; allow some time to see a slime mold expand a good distance from its source (a good spot may be the maintenance room in Ch. 1 Lvl 1). Time permitting, make it to Chapter 2, Level 1, where the calamity objects can be seen harming the slime mold and the wall-climbing mechanic can be seen, as that will fully cover the game mechanics. There isn't much need to show anything specific past Ch. 2 Lvl 1.]

During the demonstration:

- **Andy will explain the properties of slime mold that are actively in play on the screen.**
- BoWen can also explain the different tools the players have and the destructible environment mechanic, or anything else he worked on, as well as provide insight on the background story including visual and auditory design decisions.
- **Accept any ad hoc request to demonstrate a specific mechanic;** don't worry about time, we have lots of it. Make use of the "**Restart level**" button in the pause menu if needed to re-demonstrate something without restarting the whole game.

End of demonstration, back to slides - BoWen

Our project excels in usability, through our simple and intuitive control scheme, which can be easily referenced inside the game's pause menu accessed with Escape.

In addition, the performance of this software is smooth and stable across a variety of devices, and we have support for Windows, Mac, and Linux operating systems.

Lastly, through user surveys, we have seen strong positive feedback for our game's artistic style and design, and our users have successfully understood the message our game was intending to convey - the message of being one with nature, rather than against it.